

Bait Breakdown

Feature Purpose:

Bait Breakdown is a proposed PhishGuard feature that expands the Sonar Analysis Log by giving users a clear explanation for each warning detected during a URL scan. Instead of only showing labels such as “Missing HTTPS,” “URL Shortener,” or “Excessive URL Length,” this feature explains what was detected, why it may be risky, and what action the user should take next.

User Problem Addressed:

Users may not understand why a specific scan warning makes a URL suspicious. A risk score can alert users that something may be unsafe, but it does not always explain the reason behind the score. Bait Breakdown helps users understand the meaning of each warning so they can make safer decisions before opening a suspicious link.

Related Phishing Indicator:

This feature connects to phishing indicators such as suspicious links, missing HTTPS, URL shorteners, excessive URL length, redirects, phishing keywords, impersonation attempts, and requests for sensitive information.

Research Support:

The research supports this feature because several sources explain the importance of identifying and explaining phishing warning signs. CISA and the FTC support user awareness around suspicious links, phishing messages, and requests for personal or financial information. NIST supports the idea of evaluating phishing cues that affect how users detect suspicious messages. OWASP supports careful URL handling because unsafe redirects and user controlled URLs can be abused in phishing. Cloudflare’s SMS phishing case study also supports the need to explain suspicious links used in real world phishing attacks.

User Explanation:

Bait Breakdown explains each scan warning in simple language. For example, if PhishGuard detects a URL shortener, the feature could explain that shortened URLs can hide the real destination of a link. If the scan detects missing HTTPS, the feature could explain that secure websites usually use HTTPS to help protect user information. These explanations help users understand the risk without needing advanced cybersecurity knowledge.

Recommended User Action:

The user should avoid clicking suspicious links until the sender or destination can be verified. If the warning involves a shortened URL, suspicious redirect, or request for personal information, the user should not open the link or enter any sensitive information. The user should instead visit the official website directly or verify the message through a trusted contact method.

Connection to Risk Score:

Each warning explained by Bait Breakdown can help justify the final risk score. More serious

indicators, such as suspicious redirects, phishing keywords, missing HTTPS, or requests for sensitive information, may increase the risk level. This feature helps users understand why the score was assigned by connecting the score to specific warning signs found during the scan.

Prototype Status:

Planned feature improvement. The current prototype already includes a Sonar Analysis Log that displays detected warnings. Bait Breakdown would build on that section by adding clearer explanations and recommended actions for each warning.

Related User Story:

As a user, I want to understand why a URL was flagged as suspicious so that I can make a safer decision before opening it.

Acceptance Criteria:

The feature includes a warning title for each detected issue.

The feature provides a simple explanation of what was detected.

The feature explains why the warning may be risky.

The feature includes a recommended user action.

The explanation is written in clear language for non-technical users.

The warning explanation connects to the overall risk score or scan result.

The feature stays aligned with PhishGuard's scope of explainable phishing risk analysis and cybersecurity awareness.

Notes for Team Review:

The team should decide whether Bait Breakdown should appear directly inside the Sonar Analysis Log or as an expandable section under each warning. The team should also decide how much text should be shown in the prototype, so the explanations are useful without making the interface too crowded.